Dark Universe — Team Reading List & Physics Primer

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How to use this

Everyone reads the physics primer below (about 10 minutes), then two papers from their own lane. Nobody needs to read a paper end to end: skim the abstract, the main figure and the method section.

The single most useful paper for the whole team is Jeffrey et al. 2024 (DES Y3 simulation-based inference). It runs almost exactly our pipeline at professional scale: DES Y3 mass maps, peak counts and power spectra, simulation-based inference. Treat its results as the answer key; our S8 should land near its value of about 0.80.

The physics in one page

The core idea: dark matter can't be seen, but its gravity bends light from galaxies behind it, so the pattern of distorted galaxy shapes is a map of the mass.

1. **Dark matter.** About 85% of the matter in the universe emits no light. We know it's there only through gravity: galaxy rotation, cluster dynamics, the cosmic microwave background, and lensing. Over cosmic time it collapses into a "cosmic web" of dense clusters, long filaments and empty voids. Galaxies form inside this scaffolding.
2. **Gravitational lensing.** Mass curves spacetime, so light passing a mass is deflected. A huge cluster can produce visible arcs (strong lensing). Everywhere else the effect is tiny, stretching background galaxy images by about 1%. That is weak lensing.
3. **Shear and the averaging trick.** One galaxy's shape tells you nothing, because galaxies are intrinsically elliptical in random directions. Averaging the shapes of many galaxies in one patch cancels the random part and leaves the small coherent stretch, called shear. This is why it takes professional surveys measuring over 100 million galaxies.
4. **The convergence map (κ).** Shear can be mathematically inverted into convergence, κ: the projected mass along each line of sight. That inversion is the Kaiser–Squires method, essentially a Fourier-space filter. A κ map is our "dark matter map". It is a 2D projection, weighted toward mass roughly halfway between us and the source galaxies. DES released these maps, so we download rather than reconstruct.

5. **What the universe's recipe means.** In the standard model (ΛCDM), two numbers matter most for lensing. Ωm is the fraction of the universe's energy that is matter. σ8 is how clumpy matter is today, measured on 8-megaparsec scales. Lensing mostly constrains one combination of them:

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S_8 = \sigma_8 \sqrt{\Omega_m / 0.3}
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6. **Summary statistics.** A raw map is too big to compare directly with theory, so we compress it. The power spectrum measures how much structure exists at each size scale, and is the standard tool. Peak counts tally the bright spots in the map, which correspond to massive halos and clusters. Peaks catch non-Gaussian information that the power spectrum misses, because gravity makes the web lumpy, not like smooth noise.
7. **Simulation-based inference (SBI).** There's no clean formula linking peak counts to Ωm and S8. So we simulate thousands of universes with known parameters (CosmoGridV1), compute the same statistics on each, and train a small neural network (a normalizing flow) to learn the probability of parameters given statistics. Feeding in the real map's statistics then returns a posterior: best values with honest error bars. The trust check is coverage: on held-out simulations, the true answer should fall inside the 68% interval about 68% of the time.
8. **Why S8 is interesting.** Planck measured the early universe (the cosmic microwave background) and predicts how clumpy today's universe should be. Lensing measures today's clumpiness directly. For years lensing came out roughly 2σ lower: the "S8 tension". The completed KiDS survey (2025) now finds agreement with Planck, so the tension may be fading. Our demo should show where our measurement lands, not claim new physics.
9. **Galaxy mergers and environment.** Galaxies merge when they meet slowly enough for gravity to pull them together. That happens most in groups. In the cores of rich clusters, galaxies move at around 1,000 km/s, too fast to merge easily. So the expected link between merger rate and dark-matter density is real but not simply "denser means more mergers". That makes it a genuine question for our citizen-science layer, not a foregone conclusion.

Reading list

Start with the two "Read first" papers in your lane; the rest are reference for when you get stuck.

Who	Paper	Why read it	Priority
Everyone	Jeffrey et al. 2024, DES Y3 simulation-based inference	Our exact pipeline done professionally: DES Y3 maps, peaks plus power spectrum, SBI. The answer key.	Read first
Everyone	Jeffrey et al. 2021, DES Y3 mass maps	Describes the maps we download, what they look like and their limits. Great figures.	Read first
AI	Cranmer, Brehmer & Louppe 2020, The frontier of SBI	Short, readable overview of why and how simulation-based inference works.	Read first
AI	Tejero-Cantero et al. 2020, the sbi toolkit	The package we use; shows the neural posterior estimation workflow.	Read first
AI	Papamakarios et al. 2021, Normalizing flows	Reference for the network inside sbi. Skim sections 1 to 3 only.	Reference
AI	Kacprzak et al. 2023, CosmoGridV1	The training simulations: which parameters vary, file formats, resolution.	Read first (shared with CS/EE)
CS/EE	Jeffrey et al. 2020, DeepMass	Mass mapping as an image-reconstruction problem; Kaiser–Squires vs. deep learning, in signal-processing terms.	Read first
CS/EE	Kratochvil, Haiman & May 2010, peak counts	Why counting peaks in the map carries cosmological information beyond the power spectrum.	Reference
CS/EE	Kilbinger 2015, Cosmic shear review	The math of shear, convergence and the power spectrum when you need it.	Reference

Who	Paper	Why read it	Priority
Science lane	Mandelbaum 2018, Weak lensing for precision cosmology	Best review of the systematics we must not overclaim around.	Read first
Science lane	Planck 2018 cosmological parameters	Source of the comparison S8 value. Only the parameter tables are needed.	Reference
Science lane	Wright et al. 2025, KiDS-Legacy cosmic shear	Newest major result: S8 agrees with Planck. Essential for framing the tension honestly.	Read first
Science lane	Walmsley et al. 2023, Galaxy Zoo DESI	The citizen-science catalogue to use: 8.7M galaxies including DES imaging, out to redshift about 0.4.	Read first
Science lane	Lintott et al. 2008, Galaxy Zoo	The original citizen-science project; good for the pitch story.	Reference
Science lane	Darg et al. 2010, Galaxy Zoo merger fraction	How Galaxy Zoo volunteers identified mergers and what fraction of galaxies are merging.	Reference
Background	Bartelmann & Schneider 2001, Weak gravitational lensing	The classic textbook-length review. Look things up; don't read it.	Reference

Gotchas that change the build

The citizen-science layer only works if the merger galaxies sit in the same sky area and at the same distances the lensing map is sensitive to.

- **Use Galaxy Zoo DESI, not the older SDSS-based catalogues.** GZ DESI covers DES imaging and reaches redshift about 0.4. The original Galaxy Zoo and GZ DECaLS sit mostly in the northern sky with little DES overlap, and are limited to redshift below

about 0.15.

- **Match distances.** The lensing map is most sensitive to mass roughly halfway to the source galaxies, around redshift 0.3 to 0.4 for DES. Nearby merger galaxies barely register in it. Select mergers by redshift, and if possible compare against the lowest-redshift tomographic map.
- **Mind the resolution.** The released maps are smoothed on arcminute scales, coarser than individual galaxy groups. Compare merger fractions in bins of map density across many patches, not galaxy by galaxy.
- **Expect a non-simple answer.** Mergers favour groups and are suppressed in cluster cores, so a rise and then a flattening or dip with density is physically plausible. Present it as a measured relation, not a confirmation.
- **Call GZ DESI what it is.** Its labels are deep-learning predictions trained on volunteer votes, not raw votes for every galaxy. That's still citizen science, but say "trained on volunteer classifications".
- **Frame S8 carefully.** Jeffrey et al. 2024 found S8 of about 0.80 from DES Y3 maps, and KiDS-Legacy now agrees with Planck. A result near 0.80 means our pipeline works. Don't call it evidence for or against the tension.
- **Simulations must match the data.** Train on simulated maps with the same mask, noise level and smoothing as the real DES map, or the network's answer is quietly biased. The coverage check catches broken training but not this mismatch.

Sources

- Wright et al. 2025, KiDS-Legacy (arXiv) · UCL Discovery abstract
- Jeffrey et al. 2024, DES Y3 SBI (arXiv)
- Walmsley et al. 2023, Galaxy Zoo DESI (arXiv)